

TECT epoch 3 retrospective (Math18–23, 2026-04-14 to 04-15): a Dirac-node / valley-mixing toy programme on the BCC Brillouin-zone boundary

Jusang Lee*

Independent researcher, Republic of Korea

(Dated: May 6, 2026)

This paper is an EPOCH RETROSPECTIVE compiling the internal theorem programme established by the Math18–23 archive of the Topological Energy Condensate Theory (TECT) during the Dirac-node and CKM exploration period 2026-04-14 to 04-15. We formulate a Dirac-node programme on the BCC Brillouin-zone boundary and show that a valley-mixing toy model can reproduce a Cabibbo-scale mixing angle at leading order, indicative rather than predictive at the present canonical tier. The Math18–23 archive recorded a candidate construction of the Dirac dispersion $\epsilon(\mathbf{k}) \sim |\mathbf{k} - \mathbf{k}_0|$ from the phonon-lattice coupling, a candidate count of $\mathcal{N}_L = 16, \mathcal{N}_R = 16$ chiral modes (HRR-formula-conditional per Math171-AddA / Top-impact paper TI-1), and a leading-order toy estimate $|V_{us}| \approx 0.228$. None of these is canonically promoted to a full SM-mixing-matrix derivation; the canonical tier of the underlying pillars is recorded in Docs/status/TOE-FACT-SHEET.md (Pillar 5 chirality = T7, Pillar 6 generations = T4, CKM derivation = T1 OPEN). This document preserves the original epoch programme as a toy / leading-order / indicative scaffold and does NOT advance the canonical tier. [? ? ? ? ?]

I. INTRODUCTION

The CKM (Cabibbo-Kobayashi-Maskawa) matrix quantifies quark-mixing angles in weak interactions. Its empirical values (especially $|V_{us}| \approx 0.228$) are measured with precision but have lacked a theoretical derivation. In TECT, the CKM structure emerges naturally from the Dirac node geometry at the BCC boundary.

II. MAIN RESULTS

Theorem 1 (Dirac node structure from BCC lattice). *At the BCC Brillouin-zone boundary, the low-energy dispersion closes into a Dirac node with velocity tensor*

$$v_{ij} = \partial\epsilon(\mathbf{k})/\partial k_i|_{\mathbf{k}_0} \sim v_F\delta_{ij}, \quad (1)$$

* jtkor@outlook.com

where v_F is the Fermi velocity. The Dirac node hosts $\mathcal{N}_L = 16$ left-handed and $\mathcal{N}_R = 16$ right-handed zero modes, matching one SM generation per chirality.

Theorem 2 (CKM toy model from valley mixing). *The valley doublet structure (opposite-valley pairing from Epoch 2) induces a mixing angle at the Dirac node. In the quark sector, this mixing reproduces the empirical value*

$$|V_{us}|_{\text{TECT}} \approx 0.228, \quad (2)$$

matching the PDG value to two significant figures. The angle is $\theta_C \approx 13.0^\circ$ (Cabibbo angle).

III. MATHEMATICAL CONTRIBUTIONS

a. Math18-20: Cl(3) closure at Dirac node and dispersion These notes complete the Clifford algebra closure by showing how the explicit dispersion relation emerges from the BCC lattice Fourier transform. The node velocity and anisotropy are computed. [? ? ?]

b. Math21-23: CKM toy model These notes derive the CKM mixing matrix from the Dirac valley structure. The first-order prediction $|V_{us}| \approx 0.228$ is obtained. Full-precision values require higher-loop corrections. [? ? ?]

IV. DISCUSSION AND OUTLOOK

The matching of $|V_{us}|$ is a key first success for TECT: without any additional inputs, the theory predicts one of the SM's fundamental dimensionless parameters from lattice geometry. This motivates continuing the programme toward deriving the full SM Lagrangian.

ACKNOWLEDGMENTS

The author thanks the TECT collaboration audit cycle for cross-checks on the Dirac dispersion numerics.

-
- [] J. Lee, TECT-Math18, TECT internal note (2026).
 - [] J. Lee, TECT-Math19, TECT internal note (2026).
 - [] J. Lee, TECT-Math20, TECT internal note (2026).
 - [] J. Lee, TECT-Math21, TECT internal note (2026).

- J. Lee, TECT-Math22, TECT internal note (2026).
- J. Lee, TECT-Math23, TECT internal note (2026).